Q1 - 2024 (08 Apr Shift 1)

If $\sin x=-\frac{3}{5}$, where $\pi<x<\frac{3\pi}{2}$, then $80\left(\tan^2 x-\cos x\right)$ is equal to

(1) 108

(2) 109

(3) 18

(4) 19

Q2 - 2024 (08 Apr Shift 2)

If the value of $\frac{3\cos 36^{\circ}+5\sin 18^{\circ}}{5\cos 36^{\circ}-3\sin 18^{\circ}}$ is $\frac{a\sqrt{5}-b}{c}$, where a, b, c are natural numbers and $\gcd(a, c)=1$, then $a+b+c$ is equal to :

(1) 40

(2) 52

(3) 50

(4) 54

Q3 - 2024 (09 Apr Shift 1)

Let $|\cos\theta\cos(60-\theta)\cos(60+\theta)|\leq\frac{1}{8}, \theta\epsilon[0,2\pi]$. Then, the sum of all $\theta\epsilon[0,2\pi]$, where $\cos 3\theta$ attains its maximum value, is :

(1) 15π

(2) 18π

(3) 6π

(4) 9π

Answer Key

Q1 (2) **Q2** (2) **Q3** (3)

Q1

$\sin x = \frac{-3}{5}, \pi < x < \frac{3\pi}{2}$

$\tan x = \frac{3}{4} \cos x = -\frac{4}{5}$

$80\left(\tan^2 x - \cos x\right)$

$= 80\left(\frac{9}{16} + \frac{4}{5}\right) = 45 + 64 = 109$

Q2

$$\frac{\frac{3(\sqrt{5}+1)}{4} + 5\left(\frac{\sqrt{5}-1}{4}\right)}{5\left(\frac{\sqrt{5}+1}{4}\right) - 3\left(\frac{\sqrt{5}-1}{4}\right)} = \frac{8\sqrt{5}-2}{2\sqrt{5}+8}$$

$$= \frac{4\sqrt{5}-1}{\sqrt{5}+4} \times \frac{\sqrt{5}-4}{\sqrt{5}-4}$$

$$= \frac{20 - 16\sqrt{5} - \sqrt{5} + 4}{-11}$$

$$= \frac{17\sqrt{5} - 24}{11} \Rightarrow a = 17, b = 27, c = 11$$

$a + b + c = 52$

Q3

We know that

$(\cos\theta)\left(\cos(60^\circ - \theta)\right)\left(\cos(60^\circ + \theta)\right) = \frac{1}{4}\cos 3\theta$

So equation reduces to $\left|\frac{1}{4}\cos 3\theta\right| \leq \frac{1}{8}$

$\Rightarrow |\cos 3\theta| \leq \frac{1}{2}$

$\Rightarrow -\frac{1}{2} \leq \cos 3\theta \leq \frac{1}{2}$

$\Rightarrow$ maximum value of $\cos 3\theta = \frac{1}{2}$, here

$\Rightarrow 3\theta = 2n\pi \pm \frac{\pi}{3}$

$\theta = \frac{2n\pi}{3} \pm \frac{\pi}{9}$

As $\theta \in [0, 2\pi]$ possible values are

$\theta = \left\{\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}\right\}$

Whose sum is

$\frac{\pi}{9}+\frac{5\pi}{9}+\frac{7\pi}{9}+\frac{11\pi}{9}+\frac{13\pi}{9}+\frac{17\pi}{9}=\frac{54\pi}{9}=6\pi$